

SUPPLY MANAGEMENT SYSTEM — Functional Specification Document

FSD Addendum 31 — Manufacturing Flow Corrections & UX Consolidation

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Impact	3 migrations, 6 modified entities, 4 new/modified UI panels, 2 workflow changes

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1. Purpose & Scope

This addendum corrects and refines 10 specific flows introduced in FSD Addendum 30 (Manufacturing, Production & Allocation Engine). It does not add new modules — it modifies existing behavior based on business review feedback to improve usability, consolidate UX, and align the manufacturing loop with operational workflow requirements.

1.1 Design Principles

- **Consolidation over Proliferation** — BOM management is embedded in the Product form rather than a separate module. Shortage views are consolidated across all Production Orders into a single Purchase Required dashboard.
- **Draft-First Workflow** — Auto-generated Purchase Orders are created in Draft state, not auto-submitted. This gives procurement teams review and edit capability before supplier engagement.
- **Conditional Enforcement** — Sale Order quantity limits (Min/Max) only apply when explicitly configured on a product. NULL or 0 values mean no restriction.
- **Traceability** — Production Orders carry a direct reference to their originating Sale Order when demand-driven, enabling end-to-end trace from customer order through production to fulfillment.
- **Operational Consolidation** — Material shortages across all Production Orders are aggregated per product into a single Purchase Required view, preventing duplicate purchase orders for the same material.

1.2 Scope Summary

Change ID	Title	Type	Addendum 30 Ref
C1	Sale Order Quantity Limits (Min/Max)	New Feature	§6 Product Catalog
C2	Product Tab Consolidation	UX Change	§29.1 Product UI
C3	Draft Purchase Order Flow	Workflow Change	§13 Supply Requirement Engine
C4	Inline BOM on Product Form	UX Restructure	§7–9 BOM Management, §29.2 BOM UI
C5	BOM Effective Date Auto-Default	Default Change	§7.1 BillOfMaterials
C6	Remove Warehouse from BOM	Schema Change	§7.2 BillOfMaterialLines
C7	Production Order ↔ Sale Order Link	Schema Extension	§11 ProductionOrders
C8	Material Availability Tab	New UI Panel	§12 PMR, §14 Allocation
C9	Consolidated Purchase Required View	New Feature	§13 Supply Requirements
C10	GRN → Allocation → Reservation Flow	Workflow Refinement	§15, §18, §20

2. Change Matrix — Addendum 30 Impact

This section maps each change to the specific Addendum 30 sections it supersedes, modifies, or extends.

Change	ADD-030 Section	Action	Impact Description
C1	§6 Products	EXTEND	Add sale_order_min_qty, sale_order_max_qty to Products table. Add validation in Sale Order line creation.
C2	§29.1 Product UI	MODIFY	Merge Manufacturing tab content into Classification tab. Remove separate Manufacturing tab.
C3	§13 Supply Req Engine	MODIFY	Replace auto-submit + auto-send-to-supplier with Draft creation + notification. Add default supplier pre-selection with edit capability.
C4	§7-9 BOM, §29.2 BOM UI	RESTRUCTURE	Remove standalone BOM module UI. BOM is now inline on Product form for manufacturing products. Mandatory when product_type requires BOM. Two-panel layout: BOM list + BOM editor.
C5	§7.1 BillOfMaterials	MODIFY	effective_from defaults to GETDATE() instead of NULL. Applies to new BOM creation only.
C6	§7.2 BOM Lines	REMOVE	Remove warehouse_id from BillOfMaterialLines. Warehouse is determined at Production Order level, not BOM level.
C7	§11 ProductionOrders	EXTEND	Add sale_order_id, sale_order_line_id columns. Populated when PO originates from Sale Order demand. NULL for directly created POs.
C8	§12, §14 PMR + Alloc	NEW PANEL	New 'Material Availability' tab on Production Order detail. Two sections: Available (with Reserve button) and Shortage (with Allocation Request button).
C9	§13 Supply Requirements	RESTRUCTURE	Replace per-Production Order shortage view with consolidated Purchase Required dashboard. Aggregate shortages across all POs per product. Single-line per product with summed qty.
C10	§15, §18, §20	REFINE	Clarify GRN → allocation run → inventory reservation → PO material status update flow. Authorized user triggers allocation. Items move from Shortage to Available on PO.

3. C1 — Sale Order Quantity Limits (Min/Max per Product)

3.1 Overview

Add two new parameters to the Product Stock Settings: Sale Order Min Qty and Sale Order Max Qty. These enforce minimum and maximum order quantities when a Sale Order line is created or updated for this product variant.

⚠ CRITICAL: When both `sale_order_min_qty` and `sale_order_max_qty` are NULL or 0, NO quantity validation is applied. The limit is conditional — it only fires when explicitly configured.

3.2 Schema Change

Modify: `lookups.Products`

Products — New Columns (ADD)

Column	Type	Nullable	Description
sale_order_min_qty	DECIMAL(18,4)	NULL	Minimum allowed quantity per Sale Order line. NULL or 0 = no minimum.
sale_order_max_qty	DECIMAL(18,4)	NULL	Maximum allowed quantity per Sale Order line. NULL or 0 = no maximum.

3.3 Validation Logic

Applied at Sale Order line creation and update (SaleOrderLineService.Create/Update):

```
csharp
```

```
// Pseudocode – Sale Order Line Quantity Validation
var product = await _productRepo.GetByIdAsync(lineDto.ProductId);

decimal minQty = product.SaleOrderMinQty ?? 0;
decimal maxQty = product.SaleOrderMaxQty ?? 0;

if (minQty > 0 && lineDto.Quantity < minQty)
    throw new BusinessException(
        $"Quantity {lineDto.Quantity} is below minimum order quantity of {minQty}");

if (maxQty > 0 && lineDto.Quantity > maxQty)
    throw new BusinessException(
        $"Quantity {lineDto.Quantity} exceeds maximum order quantity of {maxQty}");

// Cross-validation: if both set, min must be <= max
// This is validated on Product save, not on SO line save
```

3.4 Product-Level Cross-Validation

On Product create/update, if both values are provided:

- sale_order_min_qty must be $\leq$ sale_order_max_qty (when both are > 0)
- Both values must be ≥ 0 (negative quantities not allowed)
- Values are per-line, not per-order (each SO line is validated independently)

3.5 Business Rules

Rule ID	Rule	Enforcement
BR-C1-01	Min/Max validation fires ONLY when value is non-null AND > 0	SaleOrderLineService
BR-C1-02	When min is set but max is not → only minimum enforced	SaleOrderLineService
BR-C1-03	When max is set but min is not → only maximum enforced	SaleOrderLineService
BR-C1-04	Validation applies to each SO line independently, not aggregated across lines	SaleOrderLineService
BR-C1-05	On Product save: min must be ≤ max when both are > 0	ProductService
BR-C1-06	Changing min/max on a product does NOT retroactively validate existing Sale Orders	N/A

3.6 UI Placement

Location: Product form → Stock Parameters section (existing section)

- Add 'Sale Order Min Qty' numeric input — placed below existing stock parameters
- Add 'Sale Order Max Qty' numeric input — placed below Min Qty
- Both fields accept decimal values (4 decimal places)
- Placeholder text: 'Leave empty for no limit'
- When max < min on save: inline validation error 'Maximum must be greater than or equal to minimum'

3.7 Sale Order UI Impact

When a user enters a quantity on a Sale Order line and tabs out or saves:

- If qty < min: inline error below qty field — 'Minimum order quantity for [Product Name] is [min]'
- If qty > max: inline error below qty field — 'Maximum order quantity for [Product Name] is [max]'
- Error clears when qty is corrected
- Save button is disabled while validation errors exist

4. C2 — Product Tab Consolidation (Manufacturing → Classification)

4.1 Overview

The Manufacturing tab defined in Addendum 30 (§29.1) is removed as a separate tab. All manufacturing-related fields are merged into the existing Classification tab on the Product form. This reduces tab proliferation and groups all product categorization and type information in one place.

4.2 Fields Moved to Classification Tab

The following fields from the Manufacturing tab are relocated to the Classification tab, appearing after the existing classification fields:

Field	Type	Position in Classification Tab	Visibility Condition
Product Type	Dropdown (8 values)	After existing category/sub-category fields	Always visible
Supply Method	Dropdown (4 values)	Below Product Type	Always visible
Is Manufacturable	Toggle	Below Supply Method	Always visible
Is BOM Input	Toggle	Below Is Manufacturable	Visible when Product Type = FinishedGood or SemiFinished
Is Saleable	Toggle	Below Is BOM Input	Always visible
Is Purchasable	Toggle	Below Is Saleable	Always visible
Is Stockable	Toggle	Below Is Purchasable	Always visible
Default Production Warehouse	Dropdown	Below Is Stockable	Visible when Is Manufacturable = true
Lead Time Days	Numeric	Below Default Production Warehouse	Visible when Is Manufacturable = true

4.3 Classification Tab Layout

The Classification tab is now organized into two visual sections:

- **Section 1 — Product Classification (existing):** Category, Sub-Category, Brand, Unit of Measure, HSN/Tax code, etc.
- **Section 2 — Product Type & Manufacturing (new, below Section 1):** Product Type, Supply Method, manufacturing toggles, production warehouse, lead time. This section has a subtle section header 'Manufacturing & Supply' with a horizontal divider above it.

 **NOTE:** The separate Manufacturing tab route/component is removed entirely. Any existing navigation links to it should redirect to the Classification tab.

5. C3 — Supply Requirement Engine: Draft Purchase Order Flow

5.1 Overview

Addendum 30 §13 specified that the Supply Requirement Engine auto-creates and auto-submits Purchase Orders to suppliers when a component has `supply_method=Purchase`. This is changed: the engine now creates Purchase Orders in Draft status and sends a notification to the procurement team. The user reviews, optionally changes the supplier and quantity, then manually submits.

 **CRITICAL:** The auto-submit and auto-send-to-supplier behavior from Addendum 30 §13 is fully replaced by this Draft-first workflow. No Purchase Order is sent to a supplier without human review.

5.2 Revised Flow

When `ISupplyRequirementEngine.ProcessShortages()` identifies a component with `supply_method=Purchase`:

```
// BEFORE (Addendum 30 — SUPERSEDED):  
// 1. Create PO → 2. Auto-submit → 3. Auto-send to supplier  
  
// AFTER (Addendum 31 — CURRENT):  
// 1. Create PO in Draft status  
// 2. Auto-select default supplier from product.default_supplier_id  
// 3. Set quantity from SR.required_qty (subject to user update)  
// 4. Link SR ↔ PO (sr.linked_po_id = po.purchase_order_id)  
// 5. Send notification + email to users with 'purchase_order_write' permission  
// 6. User reviews → changes supplier/qty if needed → manually submits
```

5.3 Auto-Created Purchase Order Details

PO Field	Value	Editable by User
status	Draft	No (transitions via workflow)
supplier_id	product.default_supplier_id (from Product master)	Yes — user can change before submit
quantity	supply_requirement.required_qty	Yes — user can adjust before submit
required_by_date	production_order.planned_start_date (from parent PO)	Yes
source_type	'PRODUCTION' (new marker)	No
source_reference_id	supply_requirement.supply_requirement_id	No
notes	'Auto-generated from Production Order [PO#] for [Product Name]'	Yes
created_by	SYSTEM (service account)	No

5.4 Default Supplier Selection

The system pre-selects the supplier using this priority:

1. **product.default_supplier_id** — if a default supplier is configured on the product
2. **Last supplier used** — query the most recent approved PO for this product to find the last supplier
3. **NULL** — if no supplier history exists, leave supplier_id NULL. User must select before submitting.

 **NOTE:** When supplier_id is NULL on the auto-created Draft PO, the notification message includes 'Supplier selection required' to alert the reviewer.

5.5 Notification

On Draft PO creation from Supply Requirement Engine:

- Notification type: `PurchaseOrderDraftCreated`
- Recipients: All users in the same org_id with permission claim 'purchase_order_write'
- Channel: In-app notification (auth.Notifications) + Email via Hangfire
- Email subject: `[SMS] Purchase Order [PO#] created for review – [Product Name]`

- Email body includes: PO number, product name, required quantity, required-by date, supplier (or 'Not selected'), link to PO edit page, originating Production Order number

5.6 Business Rules

Rule ID	Rule	Enforcement
BR-C3-01	Auto-created POs are ALWAYS in Draft status — never auto-submitted	SupplyRequirementEngine
BR-C3-02	Default supplier is pre-selected but always editable before submit	PO Edit UI
BR-C3-03	Quantity is pre-filled from SR but always editable before submit	PO Edit UI
BR-C3-04	PO cannot be submitted without a supplier_id	PurchaseOrderService.Submit()
BR-C3-05	The SR ↔ PO link is established at creation and immutable	SupplyRequirementEngine
BR-C3-06	If user changes quantity on PO, the SR.required_qty is NOT updated (SR reflects original demand)	PurchaseOrderService
BR-C3-07	Multiple SRs for the same product may link to the same Draft PO (consolidated purchasing)	SupplyRequirementEngine

6. C4 — Inline BOM Management on Product Form

6.1 Overview

Instead of a separate Bill of Materials module with its own navigation and pages (Addendum 30 §29.2), BOM management is integrated directly into the Product form. When a product has product_type that requires manufacturing (FinishedGood, SemiFinished, or any product with supply_method=Manufacture), a BOM section appears on the product form. BOM data entry is mandatory for such products.

⚠ CRITICAL: The standalone BOM list page and BOM create/edit pages from Addendum 30 §29.2 are REMOVED. All BOM operations happen inline on the Product form. The BOM API endpoints (§28.2) remain unchanged — only the UI entry point changes.

6.2 BOM Section Visibility

The BOM section appears on the Product form when:

- product_type IN (FinishedGood, SemiFinished) — always shows BOM section
- OR supply_method = Manufacture — always shows BOM section
- OR is_manufacturable = true — always shows BOM section

When visible, the BOM section is mandatory:

- The product cannot be saved/activated without at least one BOM in Draft or higher status
- Validation message: 'At least one Bill of Material is required for a manufacturing product'

6.3 Two-Panel Layout

The BOM section on the Product form uses a two-panel layout:

Panel 1 — BOM List (Left Panel, 35% width)

- Lists all BOMs for this product (same data as the removed standalone BOM list)
- Each row shows: BOM Number, Version, Status (badge), Effective From, Effective To, Is Default
- Status badge colors: Draft (grey), Submitted (blue), Approved (amber), Active (green), Obsolete (grey strikethrough), Rejected (red)
- Click a BOM row → loads it in Panel 2 for viewing/editing
- '+ New BOM' button at top of list → opens blank form in Panel 2
- Active BOM is visually highlighted (green left border)

Panel 2 — BOM Editor/Viewer (Right Panel, 65% width)

When a BOM is selected or new BOM is being created:

- Header: BOM Number (auto-generated on create), Version, Status badge
- Fields: Base Quantity, Base UOM, Yield Percentage, Notes
- Effective From date (auto-defaults to today — see C5)
- Effective To date (optional)
- BOM Lines table: Component Product (searchable dropdown), Quantity, UOM, Wastage %, Scrap %, Is Critical (toggle), Notes
- Add Line button at bottom of lines table
- Inline line editing — click a line to edit in-place
- Delete line: trash icon on each row (with confirmation)

Workflow action buttons (bottom of Panel 2, based on current status):

- **Draft:** [Save] [Submit for Approval] [Delete]
- **Submitted:** [Approve] [Reject] (4-eyes principle enforced)
- **Approved:** [Activate] (deactivates other Active BOMs for this product)
- **Active:** [Create New Version] [Obsolete]
- **Rejected:** [Edit & Resubmit] (creates new version in Draft)
- **Obsolete:** [Create New Version]

6.4 Empty State

When a manufacturing product has no BOMs yet:

- Panel 1 shows empty state: 'No Bill of Materials defined'
- Panel 2 shows: 'Add a Bill of Materials to define the components required to manufacture this product.' with a prominent '+ Create First BOM' button
- The Product save button shows a warning badge: 'BOM required'

6.5 BOM Version Comparison

In Panel 1, when multiple versions exist for the same product:

- A 'Compare Versions' link appears below the list
- Opens a modal with side-by-side version comparison (same as Addendum 30 §29.2)
- Highlights: added lines (green), removed lines (red), changed quantities (amber)

6.6 Navigation Change

Navigation Item	Before (ADD-030)	After (ADD-031)
Main nav: BOM Management	Exists as top-level menu	REMOVED
BOM List page	/bom-management	REMOVED — inline on Product form
BOM Create page	/bom-management/create	REMOVED — inline on Product form
BOM Edit page	/bom-management/:id/edit	REMOVED — inline on Product form
BOM API endpoints	Unchanged	Unchanged — /api/boms/* still active
Product form	Separate Manufacturing tab	Classification tab + inline BOM section

7. C5 — BOM Effective Date Auto-Default

7.1 Change

When creating a new BOM (via the inline BOM editor in the Product form), the `effective_from` field is automatically set to today's date. The user can change it but it is no longer NULL by default.

7.2 Implementation

- Frontend: On BOM create form mount, set `effective_from = new Date().toISOString().split('T')[0]`
- Backend: In `BOMService.Create()`, if `effective_from` is null → set to `DateTimeOffset.UtcNow.Date`
- Existing BOMs with NULL `effective_from` are NOT retroactively updated
- `effective_to` remains optional (NULL by default)

7.3 Schema Impact

No schema change required. The column already allows NULL. The default is applied at the application level, not the database level. This preserves backward compatibility with existing BOMs that have NULL `effective_from`.

8. C6 — Remove Warehouse from BOM

8.1 Rationale

The `warehouse_id` on `BillOfMaterialLines` (Addendum 30 §7.2) is removed. Warehouse assignment happens at the Production Order level — the `production_warehouse_id` on the Production Order determines where materials are sourced from. Having warehouse at the BOM line level creates confusion because the same BOM may be used in different warehouses depending on the Production Order.

8.2 Schema Change

Modify: `material.BillOfMaterialLines`

- **REMOVE:** `warehouse_id` BIGINT NULL FK — drop column and foreign key constraint
- The corresponding Warehouse dropdown in the BOM line editor UI is removed

8.3 Impact on PMR/BOM Explosion

In Addendum 30 §12.2 (BOM Explosion Service), the PMR `warehouse_id` was set from the BOM line warehouse or the Production Order warehouse. With this change:

- `PMR.warehouse_id` is ALWAYS set from `ProductionOrder.production_warehouse_id`
- This simplifies the BOM explosion logic — no conditional warehouse selection

8.4 Migration

Migration drops the column. Any existing data in warehouse_id on BOM lines is discarded. This is safe because:

- BOM lines are template data, not transactional — no business data is lost
- The warehouse selection moves to Production Order, which already has production_warehouse_id

9. C7 — Production Order ↔ Sale Order Reference

9.1 Overview

Add a direct reference from Production Orders to the originating Sale Order. When a Production Order is created from a Sale Order demand (via FulfillmentRequirementService), it carries the sale_order_id and sale_order_line_id. When a Production Order is created directly (manual creation), these fields remain NULL.

9.2 Schema Change

Modify: `material.ProductionOrders`

ProductionOrders — New Columns (ADD)

Column	Type	Nullable	Description
sale_order_id	BIGINT	NULL	FK → demand.SaleOrders. The originating Sale Order, if this PO was created from SO demand. NULL for directly created POs.
sale_order_line_id	BIGINT	NULL	FK → demand.SaleOrderLines. The specific SO line that drove this production demand. NULL for directly created POs.

Both columns are nullable — they are populated ONLY when the Production Order originates from a Sale Order demand chain (SO → Fulfillment Requirement → Production Order).

9.3 Population Logic

`csharp`

```
// In FulfillmentRequirementService.CreateFromSaleOrder():
// When creating a Production Order for a SO line with fulfillment_method = Manufa

var productionOrder = new ProductionOrder {
    // ... existing fields ...
    SaleOrderId = fulfillmentReq.SaleOrderId,
    SaleOrderLineId = fulfillmentReq.SaleOrderLineId,
    SourceDemandType = "SALES_ORDER",
    SourceDemandId = fulfillmentReq.SaleOrderId,
    // ...
};

// For direct creation (ProductionOrderController.Create):
// sale_order_id and sale_order_line_id remain NULL
```

9.4 UI Display

On the Production Order detail page:

- If sale_order_id is NOT NULL: Show a 'Sale Order' link field displaying the SO number as a clickable link that opens the Sale Order detail page. Also show the SO line number.
- If sale_order_id IS NULL: Show 'Sale Order: — (Directly Created)' as a greyed-out text.
- The Sale Order field is read-only — it is set by the system, not editable by users.

9.5 Reverse Navigation

On the Sale Order detail page:

- If any Production Orders reference this Sale Order: Show a 'Production Orders' section listing all linked POs with their number, product, status, and planned qty.
- This enables end-to-end traceability: Sale Order → Production Order → Material Issue → QI → FGR → Delivery.

10. C8 — Material Availability Tab on Production Order

10.1 Overview

Add a new 'Material Availability' tab on the Production Order detail page. This tab replaces the previous inline PMR shortage display with a structured two-section layout: Material Available and Material Shortage. Each section has actionable buttons for reservation and allocation requests.

10.2 Tab Structure

Production Order Detail → Tab: 'Material Availability'

Section 1 — Material Available

Lists all materials (PMRs) that have sufficient stock in the warehouse:

Column	Description
Product	Component product name + variant
Required Qty	pmr.gross_required_qty
Available Qty	Available stock from AllocationEngine.GetAvailabilityAsync()
Warehouse	Warehouse name (from PO production_warehouse_id)
Status	Badge: Available (green)
Action	'Reserve' button — reserves this material for this Production Order

Reserve Button Behavior:

- Calls `IStockReservationService.Reserve(variantId, warehouseId, qty, sourceType=PRODUCTION_ORDER, sourceId=productionOrderId)`
- On success: row status changes to 'Reserved' (blue badge), Reserve button becomes 'Reserved' (disabled)
- On failure (insufficient stock due to concurrent allocation): shows error toast 'Stock no longer available — another allocation consumed this stock'
- Bulk action: 'Reserve All Available' button at section header — reserves all items in one batch

Section 2 — Material Shortage

Lists all materials (PMRs) with shortage_qty > 0:

Column	Description
Product	Component product name + variant
Required Qty	pmr.gross_required_qty
Issued Qty	pmr.issued_qty (what has been issued so far)
Shortage Qty	pmr.shortage_qty
Status	Badge: Shortage (red), Partial (amber)
Action	'Allocation Request' button — opens Register Demand dialog

10.3 Allocation Request Dialog

When user clicks 'Allocation Request' on a shortage line:

A modal dialog titled '**Register Demand — Allocation Request**' opens with:

Field	Pre-filled Value	Editable
Product	Component product name (from PMR)	No
Variant	Component variant (from PMR)	No
Quantity	pmr.shortage_qty	Yes (can reduce, cannot exceed shortage)
Demand Type	PRODUCTION_ORDER	No
Demand ID	production_order_id	No
Priority	Production Order priority (1-10)	Yes
Warehouse	PO production_warehouse_id	No
Required By	PO planned_start_date	Yes
Notes	Empty	Yes

On Submit:

- Calls `IAallocationEngine.AllocateAsync(variantId, warehouseId, PRODUCTION_ORDER, productionOrderId, qty, priority)`

- If stock becomes available (e.g., from GRN): allocation is fulfilled, item moves from Shortage to Available
 - If no stock available: allocation record is created as pending demand — will be fulfilled when stock arrives
 - The button label changes to 'Allocation Requested' (disabled) with a pending badge
-

11. C9 — Consolidated Purchase Required View

11.1 Overview

Instead of viewing material shortages per-Production Order (which leads to duplicate purchase orders for the same material), this change introduces a Consolidated Purchase Required dashboard. It aggregates all shortages across all active Production Orders per product into a single line item.

 **CRITICAL:** This replaces the per-Production Order 'Create PO from Shortage' button from Addendum 30. The consolidated view prevents procurement from creating separate POs for the same product that is short across multiple Production Orders.

11.2 Data Aggregation Logic

```
sql
```

```

-- Consolidated Purchase Required Query
SELECT
    pmr.component_product_id,
    p.product_name,
    p.sku,
    SUM(pmr.shortage_qty) AS total_shortage_qty,
    COUNT(DISTINCT pmr.production_order_id) AS affected_po_count,
    MIN(po.planned_start_date) AS earliest_required_date,
    p.default_supplier_id,
    s.supplier_name AS default_supplier_name
FROM material.ProductionMaterialRequirements pmr
JOIN material.ProductionOrders po ON pmr.production_order_id = po.production_order_id
JOIN lookups.Products p ON pmr.component_product_id = p.product_id
LEFT JOIN suppliers.Suppliers s ON p.default_supplier_id = s.supplier_id
WHERE pmr.shortage_qty > 0
    AND pmr.is_deleted = 0
    AND po.status IN ('Planned', 'MaterialPending', 'Ready', 'InProgress')
    AND po.org_id = @orgId
GROUP BY pmr.component_product_id, p.product_name, p.sku,
    p.default_supplier_id, s.supplier_name

```

11.3 Purchase Required Dashboard

Navigation: Manufacturing → Purchase Required (new top-level menu item)

Column	Description
Product	Product name + SKU
Total Shortage	Sum of shortage_qty across all Production Orders for this product
Affected POs	Count of Production Orders with shortage for this product (clickable — shows PO list)
Earliest Required By	Earliest planned_start_date among affected POs
Default Supplier	Product's default supplier name (or 'Not Set')
Current Stock	Current on_hand_qty for this product across all warehouses
Pending POs	Existing Draft/Submitted POs for this product (to prevent duplicates)
Action	'Create Purchase Order' button

11.4 'Create Purchase Order' Flow

When user clicks 'Create Purchase Order' on a consolidated shortage line:

- Purchase Order creation screen opens (existing PO create form from procurement module)
- Pre-filled fields:
 - Supplier: product.default_supplier_id (editable — user can change)
 - Product/variant: from the shortage line
 - Quantity: total_shortage_qty (editable — user can adjust up or down)
 - Required By: earliest_required_date from aggregation
 - Notes: 'Purchase for production shortage — [X] Production Orders affected'
 - Source Type: 'PRODUCTION' (read-only)
- PO is created in Draft status (consistent with C3)
- User reviews and submits when ready

11.5 Duplicate Prevention

Before showing the 'Create Purchase Order' button, the system checks:

- If a Draft or Submitted PO already exists for this product with source_type='PRODUCTION': show 'PO [number] already in Draft' link instead of the create button

- If the existing Draft PO quantity < total_shortage_qty: show 'PO [number] in Draft (qty: X, shortage: Y)' with a 'Update Quantity' button that opens the existing PO for editing
- This prevents procurement from accidentally creating duplicate POs for the same shortage

11.6 Affected POs Detail Drawer

Clicking the 'Affected POs' count opens a side drawer showing:

- List of Production Orders with shortage for this product
- Each row: PO Number (link), Status, Planned Qty, Shortage Qty, Planned Start Date
- Helps procurement understand the demand distribution before creating a PO

11.7 API Endpoint

New endpoint:

- `GET /api/purchase-required` — returns aggregated shortage data per product
 - Params: `?supplierId`, `?minShortageQty`, `?sortBy=shortage|urgency|product`
 - Response includes: product details, total_shortage_qty, affected_po_count, earliest_required_date, default_supplier, current_stock, existing_draft_po (if any)
-

12. C10 — GRN → Allocation → Reservation → Production Readiness Flow

12.1 Overview

This change clarifies and formalizes the end-to-end flow from Goods Receipt Note (GRN) through to Production Order material readiness. Addendum 30 described pieces of this flow across §15, §18, and §20 — this consolidates and sequences the steps explicitly, with a manual allocation trigger by an authorized user.

12.2 Complete Flow Sequence

Step-by-step sequence when a Purchase Order's goods are received:

STEP 1: GRN Received & Approved

- Inventory: stock_transactions created (GOODS_RECEIPT)
- Inventory: stock_balance.on_hand_qty increased
- Supply Requirement: IF sr.linked_po_id = this PO
 - Update sr.fulfilled_qty += grn_line.received_qty
 - Update sr.status (Open → PartiallyFulfilled → Fulfilled)

STEP 2: Authorized User Runs Allocation (MANUAL TRIGGER)

- User navigates to Allocation dashboard or Material Availability tab
- Clicks 'Run Allocation' button
- IAllocationEngine processes all pending demands for the received products
- Priority-based allocation: highest-priority demand gets stock first
- Creates AllocationRecords (status=Active) for fulfilled demands

STEP 3: Inventory Reservation

- For each fulfilled AllocationRecord where demand_type = PRODUCTION_ORDER:
 - IStockReservationService.Reserve(variantId, warehouseId, allocatedQty, sourceType=PRODUCTION_ORDER, sourceId=productionOrderId)
 - stock_balance.reserved_qty increased
 - stock_balance.available_qty decreased

STEP 4: Production Order Material Status Update

- For each affected Production Order:
 - IMaterialReadinessService.Recalculate(productionOrderId)
 - PMR items with fulfilled allocation:
 - pmr.shortage_qty decreased (or zeroed)
 - pmr.status → FullyIssued or PartiallyIssued
 - On Production Order 'Material Availability' tab:
 - Item moves from 'Shortage' section to 'Available' section
 - IF all PMRs have zero shortage:
 - PO status: MaterialPending → Ready
 - Notification: ProductionOrderReady sent
 - ELSE: PO remains MaterialPending (partial fulfillment)

STEP 5: Production Proceeds

- PO in Ready status can be started (IProductionExecutionService.Start)
- Material Issue can proceed for available materials

12.3 Manual Allocation Trigger

The allocation step (Step 2) is a manual action by an authorized user — it is NOT automatic on GRN approval. This gives the allocation team control over when and how to distribute

received stock.

- Required permission: `allocation_write`
- Location 1: Allocation Dashboard → 'Run Allocation' button (processes all pending demands)
- Location 2: Production Order → Material Availability tab → 'Check Availability' button (re-checks stock for this PO only)
- Location 3: After GRN approval, a banner on the GRN detail page shows: 'Stock received. Run allocation to distribute to pending demands.' with a 'Run Allocation' button.

12.4 Automatic vs. Manual Allocation

Scenario	Trigger	Scope
GRN Approved	Manual — user clicks 'Run Allocation'	All pending demands for received products
FGR Confirmed	Automatic — via FGRConfirmedEvent MediatR handler	Pending SO demands for the produced FG product
Manual Allocation Request (C8)	Creates pending demand	Fulfilled when stock becomes available + allocation runs
Allocation Dashboard refresh	Manual — user clicks 'Run Allocation'	All pending demands across all products

 **NOTE:** The only automatic allocation trigger is FGR Confirmation (finished goods to sales order demand). All other allocation runs require manual user action. This prevents GRN receipts from automatically consuming stock that may be needed for higher-priority demands not yet entered into the system.

12.5 Material Availability Tab State Transitions

How items move between sections on the Production Order 'Material Availability' tab:

Event	Shortage Section	Available Section
BOM Explosion (Plan)	All PMRs with shortage appear here	PMRs with existing stock appear here
Allocation Request submitted	Item shows 'Allocation Requested' badge	No change
GRN received + Allocation run	Item removed (shortage resolved)	Item appears with 'Available' status
Partial GRN + Allocation run	Shortage qty reduced, item remains	Partial qty appears in Available
Material Reserved	No change	Status changes to 'Reserved' (blue)
Material Issued	No change	Status changes to 'Issued' (green), qty updated

13. Database Migrations

Three migrations are required for this addendum. They are additive and non-breaking.

13.1 Migration Order

#	Migration Name	Schema	Type	Description
M1	AddSaleOrderQtyLimitsToProducts	lookups	ALTER TABLE	Add sale_order_min_qty DECIMAL(18,4) NULL, sale_order_max_qty DECIMAL(18,4) NULL to lookups.Products
M2	AddSaleOrderRefToProductionOrders	material	ALTER TABLE	Add sale_order_id BIGINT NULL FK, sale_order_line_id BIGINT NULL FK to material.ProductionOrders
M3	RemoveWarehouseFromBOMLines	material	ALTER TABLE	Drop warehouse_id column and FK from material.BillOfMaterialLines

13.2 Migration Details

M1 — AddSaleOrderQtyLimitsToProducts

sql

```
ALTER TABLE lookups.Products
ADD sale_order_min_qty DECIMAL(18,4) NULL;

ALTER TABLE lookups.Products
ADD sale_order_max_qty DECIMAL(18,4) NULL;

-- No data migration needed – NULL means no limit (backward compatible)
```

M2 — AddSaleOrderRefToProductionOrders

sql

```
ALTER TABLE material.ProductionOrders
ADD sale_order_id BIGINT NULL
    CONSTRAINT FK_ProductionOrders_SaleOrders
    FOREIGN KEY REFERENCES demand.SaleOrders(sale_order_id);

ALTER TABLE material.ProductionOrders
ADD sale_order_line_id BIGINT NULL
    CONSTRAINT FK_ProductionOrders_SaleOrderLines
    FOREIGN KEY REFERENCES demand.SaleOrderLines(sale_order_line_id);

CREATE INDEX IX_ProductionOrders_SaleOrder
ON material.ProductionOrders(sale_order_id)
WHERE sale_order_id IS NOT NULL;
```

M3 — RemoveWarehouseFromBOMLines

sql

```

ALTER TABLE material.BillOfMaterialLines
DROP CONSTRAINT IF EXISTS FK_BOMLines_Warehouse;

ALTER TABLE material.BillOfMaterialLines
DROP COLUMN IF EXISTS warehouse_id;

-- Safe: BOM lines are template data. Warehouse is
-- determined at Production Order level.

```

13.3 Rollback

- **M1:** DROP COLUMN sale_order_min_qty, sale_order_max_qty from Products
- **M2:** DROP COLUMN sale_order_id, sale_order_line_id from ProductionOrders (after clearing FK)
- **M3:** ADD COLUMN warehouse_id BIGINT NULL to BillOfMaterialLines + FK (data is lost but BOM lines are templates, not transactions)

14. API Changes

14.1 Modified Endpoints

Endpoint	Change	Change ID
GET/POST/PUT /api/products	Add sale_order_min_qty, sale_order_max_qty to DTOs	C1
POST/PUT /api/sale-orders/lines	Add min/max qty validation in request pipeline	C1
POST /api/boms	Default effective_from to today if null	C5
POST /api/boms/lines	Remove warehouse_id from CreateBOMLineDto	C6
GET /api/boms/lines	Remove warehouse_id from BOMLineDto response	C6
POST /api/production-orders	Add sale_order_id, sale_order_line_id to response DTO	C7
GET /api/production-orders/{id}	Include sale_order_id, sale_order_line_id + SO number in response	C7

14.2 New Endpoints

Endpoint	Method	Description	Change ID
/api/purchase-required	GET	Consolidated purchase required list — aggregated shortage per product	C9
/api/purchase-required/{productId}/affected-orders	GET	List of Production Orders with shortage for a specific product	C9
/api/production-orders/{id}/material-availability	GET	Material availability data: available + shortage sections	C8
/api/production-orders/{id}/material-availability/reserve	POST	Reserve available material for this Production Order	C8
/api/production-orders/{id}/material-availability/reserve-all	POST	Batch reserve all available materials	C8
/api/allocation/run	POST	Manual allocation trigger — processes all pending demands	C10
/api/allocation/run?productionOrderId={id}	POST	Scoped allocation for a specific Production Order	C10

14.3 Removed/Changed Behavior

Previous Behavior	New Behavior	Change ID
SupplyRequirementEngine auto-submits PO	Creates PO in Draft + sends notification	C3
BOM lines include warehouse_id	warehouse_id removed from BOM line DTOs	C6
Per-PO shortage → Create PO button	Consolidated Purchase Required dashboard	C9
GRN → Auto allocation	GRN → Manual allocation trigger	C10

15. UI Wireframes & Specifications

15.1 Product Form — Classification Tab (C2)

Product: [Product Name] [Save] [×]

[General] [Classification] [Stock] [Pricing] [BOM]

▲ active ▲ new

Product Classification

Category ▼

Sub-Category ▼

Brand ▼

UOM ▼

— Manufacturing & Supply —

Product Type ▼

Supply Method ▼

☐ Is Manufacturable

☐ Is BOM Input

☐ Is Saleable

☐ Is Purchasable

☐ Is Stockable

Default Prod WH ▼

Lead Time

15.2 Product Form — Stock Parameters (C1)

General

Classification

Stock

Pricing

BOM

▲ active

Stock Parameters

Reorder Level

Reorder Qty

Safety Stock

Max Stock Lvl

Sale Order Limits

Sale Order Min Qty

Leave empty = no limit

Sale Order Max Qty

Leave empty = no limit

15.3 Product Form — Inline BOM (C4)

15.4 Production Order — Material Availability Tab (C8)

Production Order: PROD-00085

Sale Order: SO-1042 (line 3)

Status: MaterialPending

← C7 link

[Overview]

[Material Availability]

[Issues]

[Timeline]

▲ active

Material Available

[Reserve All Available]

Product	Reqd	Avail	Warehouse	Status	Action
Steel Bar	200	200	Main WH	● Avail	[Rsrv]
Bolt M6	1200	1200	Main WH	● Avail	[Rsrv]

Material Shortage

Product	Reqd	Issued	Short	Status	Action
Paint	50	0	50	● Short	[Alloc Req]
Rubber Seal	100	40	60	● Partial	[Alloc Req]

15.5 Consolidated Purchase Required Dashboard (C9)

Manufacturing → Purchase Required

Filter: [Supplier ▼] [Min Shortage ▼] [Sort: Urgency ▼]

Product	Shortage	POs	Earliest	Supplier	Action
Paint	350	5	Sep 29	ColorCo	[Put PO]
Rubber	180	3	Oct 02	RubberLtd	[Put PO]
Epoxy	90	2	Oct 05	ChemCorp	[Put PO]
Steel Rod	50	1	Oct 10	SteelInc	PO-0892 ↑ already Draft

Clicking [Put PO] opens PO create form pre-filled.

Clicking PO-0892 opens the existing Draft PO for review.

16. Business Rules

Consolidated business rules for all 10 changes.

Rule ID	Change	Rule Description	Enforcement Point
BR-C1-01	C1	Sale Order Min/Max validation fires ONLY when value is non-null AND > 0	SaleOrderLineService
BR-C1-02	C1	When min is set but max is not → only minimum enforced	SaleOrderLineService
BR-C1-03	C1	When max is set but min is not → only maximum enforced	SaleOrderLineService
BR-C1-04	C1	Validation applies to each SO line independently	SaleOrderLineService
BR-C1-05	C1	On Product save: min must be <= max when both > 0	ProductService
BR-C1-06	C1	Changing min/max does NOT retroactively validate existing SOs	N/A
BR-C3-01	C3	Auto-created POs are ALWAYS Draft — never auto-submitted	SupplyRequirementEngine
BR-C3-02	C3	Default supplier is pre-selected but editable before submit	PO Edit UI
BR-C3-03	C3	Quantity is pre-filled from SR but editable before submit	PO Edit UI
BR-C3-04	C3	PO cannot be submitted without supplier_id	PurchaseOrderService
BR-C3-05	C3	SR ↔ PO link is established at creation and immutable	SupplyRequirementEngine
BR-C3-06	C3	Changing PO qty does NOT update SR.required_qty	PurchaseOrderService
BR-C3-07	C3	Multiple SRs for same product may link to same Draft PO	SupplyRequirementEngine

Rule ID	Change	Rule Description	Enforcement Point
BR-C4-01	C4	Manufacturing products MUST have at least one BOM	ProductService
BR-C4-02	C4	BOM section visible when product_type IN (FG, SF) OR supply_method=Manufacture	Product UI
BR-C6-01	C6	PMR.warehouse_id ALWAYS comes from ProductionOrder.production_warehouse_id	BOMExplosionService
BR-C7-01	C7	sale_order_id populated only for demand-driven POs, NULL for direct creation	FulfillmentReqService
BR-C8-01	C8	Reserve button calls IStockReservationService with PRODUCTION_ORDER source	Material Availability UI
BR-C8-02	C8	Allocation Request creates pending demand in AllocationEngine	Allocation Request dialog
BR-C9-01	C9	Purchase Required aggregates shortages across ALL active POs per product	PurchaseRequiredService
BR-C9-02	C9	Duplicate PO prevention: shows existing Draft PO instead of create button	Purchase Required UI
BR-C10-01	C10	GRN → Allocation is MANUAL (authorized user triggers)	Allocation Dashboard
BR-C10-02	C10	FGR → Allocation is AUTOMATIC (MediatR event handler)	FGRConfirmedEventHandler
BR-C10-03	C10	After allocation: items move from Shortage to Available on PO	MaterialReadinessService
BR-C10-04	C10	PO transitions MaterialPending → Ready only when ALL PMR shortages = 0	MaterialReadinessService

17. Test Scenarios

Test ID	Change	Scenario	Expected Result
T-C1-01	C1	Create SO line with qty below product min	Validation error: qty below minimum
T-C1-02	C1	Create SO line with qty above product max	Validation error: qty above maximum
T-C1-03	C1	Create SO line with qty within min-max range	SO line created successfully
T-C1-04	C1	Create SO line when min/max are both NULL	No validation — any qty accepted
T-C1-05	C1	Create SO line when min/max are both 0	No validation — any qty accepted
T-C1-06	C1	Save product with min > max (both > 0)	Validation error on product save
T-C1-07	C1	Create SO line when only min is set (max NULL)	Only min validated
T-C1-08	C1	Create SO line when only max is set (min NULL)	Only max validated
T-C3-01	C3	SR Engine creates PO for Purchase component	PO created in Draft, notification sent
T-C3-02	C3	Auto-created PO has default supplier	supplier_id = product.default_supplier_id
T-C3-03	C3	User changes supplier on Draft PO	Supplier updated, PO remains Draft
T-C3-04	C3	User changes quantity on Draft PO	Quantity updated, SR.required_qty unchanged
T-C3-05	C3	Submit PO without supplier	Validation error: supplier required

Test ID	Change	Scenario	Expected Result
T-C4-01	C4	Save FinishedGood product without BOM	Validation error: BOM required
T-C4-02	C4	BOM section visible for Manufacture product	BOM panels shown on product form
T-C4-03	C4	BOM CRUD via inline editor	All BOM operations work inline
T-C5-01	C5	Create new BOM	effective_from defaults to today
T-C6-01	C6	Create BOM line	No warehouse field present
T-C6-02	C6	BOM explosion	PMR warehouse = PO production_warehouse_id
T-C7-01	C7	PO created from SO demand	sale_order_id populated, link shown
T-C7-02	C7	PO created directly	sale_order_id NULL, shows 'Directly Created'
T-C8-01	C8	Reserve available material on PO	Stock reserved, status = Reserved
T-C8-02	C8	Allocation Request on shortage	Pending demand created in AllocationEngine
T-C8-03	C8	Reserve All Available button	All available items reserved in batch
T-C9-01	C9	View Purchase Required dashboard	Aggregated per product, summed shortage
T-C9-02	C9	Create PO from Purchase Required	PO form opens with pre-filled data
T-C9-03	C9	Draft PO already exists for product	'PO in Draft' link shown instead of create
T-C10-01	C10	GRN approved → manual allocation	Stock available, user triggers allocation

Test ID	Change	Scenario	Expected Result
T-C10-02	C10	Allocation run after GRN	Shortage items move to Available on PO
T-C10-03	C10	All shortages resolved after allocation	PO status → Ready
T-C10-04	C10	Partial fulfillment	Some items move, PO stays MaterialPending
T-C10-05	C10	FGR → automatic allocation to SO	FG stock allocated to Sales Order demand

18. Development Phases

18.1 Phase Breakdown

Phase	Changes	Estimated Days	Dependencies
Phase A — Schema & Foundation	C1, C5, C6, C7 (migrations + entity updates)	3 days	Addendum 30 Phase 1 complete
Phase B — Product UI Consolidation	C2, C4 (tab merge + inline BOM)	5 days	Phase A
Phase C — Workflow Changes	C3 (Draft PO flow + notifications)	3 days	Addendum 30 Phase 3 complete
Phase D — Material Availability & Allocation	C8, C10 (availability tab + GRN flow)	5 days	Phase C
Phase E — Consolidated Purchase Required	C9 (dashboard + aggregation)	4 days	Phase D
Phase F — Testing & Integration	All changes — integration tests	3 days	Phase A-E
TOTAL		23 days	

18.2 Parallelization Opportunities

- Phase A + Phase C can run in parallel (schema changes are independent of workflow changes)
- Phase B can start after Phase A completes
- Phase D depends on Phase C (Draft PO flow must be in place)
- Phase E depends on Phase D (allocation flow feeds Purchase Required)
- With parallelization: critical path = A → B + (C → D → E) → F = ~18 dev days

18.3 Risk Considerations

Risk	Impact	Mitigation
Inline BOM editor complexity (C4)	Medium — two-panel layout with real-time validation	Reuse existing BOM API, only change UI entry point
Consolidated Purchase Required data performance (C9)	Low — aggregate query across PMRs	Indexed views or materialized summary table if needed
Manual vs automatic allocation confusion (C10)	Medium — users may expect GRN to auto-allocate	Clear UI messaging + 'Run Allocation' banner on GRN page
Removing warehouse from BOM (C6) breaks existing data	Low — BOM lines are templates	Migration safely drops column; no transactional data affected